

Customer Requirements Document (CRD)

Project Name	Smart EV Battery Charging & Diagnostics System
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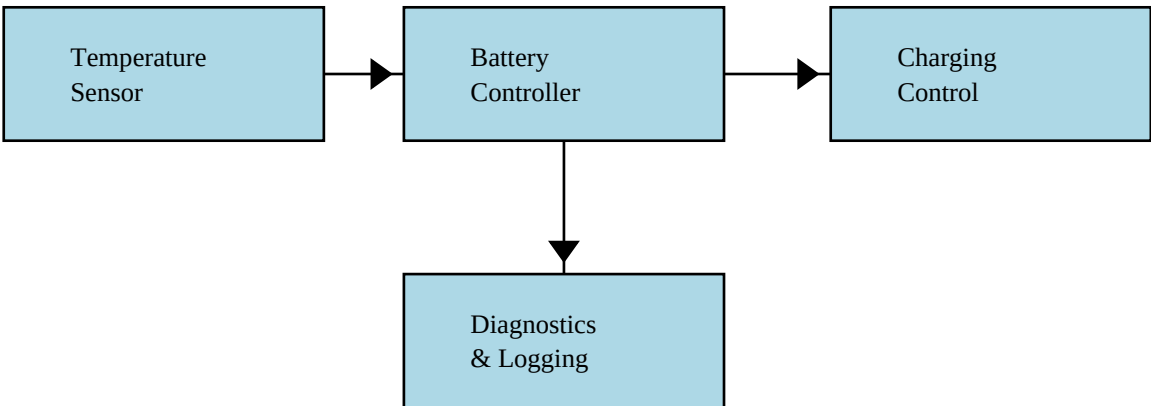
1. Introduction

The Smart EV Battery Charging & Diagnostics System is a simulated embedded software system intended to monitor battery conditions, control charging operations, detect fault conditions, and support testing and validation activities. The software is designed to follow an SDLC-based development process including requirements engineering, design, implementation, testing, and reporting. This document provides customer-level expectations and operational behavior that the engineering team shall use to derive software requirements and technical implementation details.

2. System Overview

The system continuously monitors battery temperature and charging status during EV charging operation. When unsafe conditions are detected, the software shall disable charging and generate fault information. The software architecture is expected to support: Battery temperature monitoring Charging enable/disable control Fault detection and diagnostics Logging and status monitoring Future CAN communication support Unit testing and integration testing activities Automation support for validation workflows

3. High-Level Functional Block Diagram



4. System Operational Behavior

The expected operational flow of the system is as follows: System starts in IDLE state. Battery parameters are continuously monitored. Charging is enabled when temperature remains within safe operating range. If temperature exceeds the safety threshold, the system enters FAULT state. Charging is disabled during fault conditions. Fault information is logged and diagnostic status is updated. When temperature returns to safe range, the system may recover automatically.

5. Customer Requirements

CRD ID	Customer Requirement Description
CRD_001	The system should monitor battery temperature continuously during charging operation.
CRD_002	Charging should stop automatically if battery temperature becomes unsafe.
CRD_003	The system should indicate whether charging is active or stopped.
CRD_004	Fault conditions should be detected and stored.
CRD_005	The system should support future CAN communication integration.
CRD_006	The system should maintain logs for important events and failures.
CRD_007	The response to critical faults should be quick enough to avoid battery damage.
CRD_008	The software should be modular and easy to test.
CRD_009	Engineers should be able to perform unit testing and integration testing easily.
CRD_010	The system should recover automatically when temperature returns to safe range.
CRD_011	Invalid sensor values should be handled safely.
CRD_012	The software should support automation for testing activities.

6. Engineering Expectation

The software engineering team shall analyze these customer requirements and derive detailed Software Requirements Specification (SRS) entries, design artifacts, test plans, and validation activities based on SDLC and requirement traceability principles.